

- No. of Printed Pages : 4
Roll No.

4th Semester/ Mechatronics
Subject: Embedded Systems

M.M. : 100

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is a non-vectored input? (CO4)
- a) TRAP
 - b) RST-7.5
 - c) RST-6.5
 - d) INTR
- Q.2 Which of the following statements are true for von Neumann architecture? (CO1)
- a) Separate bus between the program memory and data memory
 - b) External bus for program memory and data memory
 - c) External bus for data memory only
 - d) Shared bus between the program memory and data memory
- Q.3 The internal RAM memory of the 8051 is: (CO4)
- a) 32 bytes
 - b) 64 bytes
 - c) 128 bytes
 - d) 256 bytes
- Q.4 Which of the following is true about micro processors? (CO1)
- a) It has an internal memory
 - b) It has interfacing circuits

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain the architecture of 8085 microprocessor with the help of neat and clean block diagram. (CO4)
- Q.37 Explain with diagram inter facing of 8051 with DC motor and Relay. (CO6)
- Q.38 Illustrate the pin configuration of 8051 Microcontroller. Describe the function of each pin in detail. (CO4)

- c) It contains ALU, CU and registers
d) It uses Harvard architecture
- Q.5 If we push data onto the stack then the stack pointer (CO3)
- a) Increase with every push
b) Decreases with every push
c) Increases & decreases with every push
d) None of the mentioned
- Q.6 Accumulator is used to _____. (CO2)
- a) Stores a number
b) Add two numbers
c) Adds and stores two numbers
d) None of the above
- Q.7 What is the function of the TMOD register? (CO5)
- a) TMOD register is used to set various operation modes of timer counter.
b) TMOD register is used to load the count of the timer
c) Is the Destination or the final register where the result is obtained after the operation of the timer.
d) Is used to interrupt the timer
- Q.8 Which of the following register is not used in opcode fetch operations? (CO3)
- a) Program counter
b) Memory address register
c) Memory data register
d) Flag register
- Q.9 The Mnemonics used in writing a program is called _____. (CO5)
- a) Assembly language b) Machine language
c) High level language d) Object program

SECTION-B

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|------|--|-------|
| Q.11 | Write the use of Bus. | (CO6) |
| Q.12 | What is “Maskable interrupt”? | (CO1) |
| Q.13 | List different types of interrupts used in 8051. | (CO4) |
| Q.14 | Write the full form of PSW. | (CO2) |
| Q.15 | What is address bus? | (CO2) |
| Q.16 | What is the programme counter register. | (CO3) |
| Q.17 | What is the use of stack pointer? | (CO2) |
| Q.18 | Which interrupt has highest priority? | (CO4) |
| Q.19 | What is Debugger. | (CO1) |
| Q.20 | Write the full form of SIM. | (CO1) |

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.22 What is the difference between and assembly language. (CO2)
- Q.23 List the steps of interfacing different peripheral with 8051. (CO6)